
Towards One-Step Flow Matching: A Survey

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Abstract

Flow matching has emerged as a powerful simulation-free framework for continuous generative modeling, yet producing a single high-quality sample still requires tens to hundreds of neural function evaluations. This survey traces the rapidly evolving push toward one-step flow matching: the distillation of an entire generative trajectory into a single network call. We begin by reviewing the foundations of flow matching and conditional flow matching, then examine traditional acceleration strategies including cache-based sampling and step distillation. We next discuss closely related paradigms such as consistency models and flow-map distillation, which provide the conceptual stepping stones for direct noise-to-data transport. The core of the survey is devoted to four modern one-step families—consistency flow matching, mean flows, terminal velocity matching, and flow-map-based methods—analyzing their training objectives, theoretical guarantees, and practical trade-offs. We further summarize experimental setups, standard baselines, and state-of-the-art results, showing that the quality gap between one-step and many-step generation has narrowed substantially. By unifying these developments under a single narrative, we highlight the field’s progression from simulation-based sampling to direct prediction of the transport map, and provide practical guidance for researchers entering efficient generative modeling.

1 Introduction

Generative modeling has undergone a profound transformation with the advent of diffusion models and their continuous-time counterparts, flow-based models [Ho et al., 2020, Song et al., 2021, Lipman et al., 2023]. These approaches construct a transport between a simple noise distribution and a complex data distribution by learning a time-dependent velocity field that governs the evolution of probability mass through an ordinary differential equation (ODE). While diffusion models have achieved state-of-the-art results in image synthesis, audio generation, and scientific simulation, they share a critical limitation: generating a single sample requires simulating the ODE through many time steps, each demanding a forward pass of a large neural network. For state-of-the-art Flow Matching models, this can translate to 50–1000 function evaluations (NFEs), rendering real-time or interactive deployment impractical.

Flow Matching (FM) and its most common variant, Conditional Flow Matching (CFM) [Lipman et al., 2023], provide an especially elegant formulation of this family. By regressing vector fields of fixed conditional probability paths, FM offers a simulation-free training objective for continuous normalizing flows and yields straight, optimal-transport trajectories that are faster to integrate than the curved paths typical of diffusion models. Beyond images, FM and CFM have found broad application



Figure 1: One-step samples from Terminal Velocity Matching on both ImageNet-512 $\times$ 512 and ImageNet-256 $\times$ 256 [Zhou et al., 2026].

in speech synthesis, where they power neural vocoders and text-to-speech systems by mapping latent noise directly to raw waveforms or mel-spectrogram features; they are also used in music generation, video synthesis, protein design, and molecule generation. These successes make flow matching a natural and timely lens through which to study efficient generative modeling.

This computational bottleneck has catalyzed extensive research toward one-step and few-step generative models. The central question is whether it is possible to distill the expensive many-step sampling process into a direct map that transports noise to data in a single neural network evaluation, while preserving the quality, diversity, and controllability of the original model. Early approaches to this problem, such as caching intermediate computations, progressive distillation [Salimans and Ho, 2022], and consistency models [Song et al., 2023], provided proof-of-concept results but struggled with scalability, multi-step degradation, or the need for pre-trained teachers.

More recently, flow matching has provided a unified lens for understanding these acceleration efforts. It frames generative modeling as learning a probability flow ODE, and one-step methods can be seen as different strategies for bypassing the costly simulation of that ODE—whether by enforcing consistency along trajectories, predicting average velocities, matching terminal dynamics, or learning the flow map that jumps directly between time points. Despite their differing objectives, these approaches share a common aim: replacing iterative sampling with direct, single-step or few-steps prediction.

This survey traces the development of one-step flow matching from its conceptual origins to the current state of the art. Our contributions are as follows:

- We provide a unified narrative that connects flow matching, traditional acceleration techniques, consistency models, and modern distillation methods within a single mathematical framework.
- We analyze four key methodological advances—consistency flow matching, mean flows, terminal velocity matching, and flow-map-based methods—comparing their training requirements, theoretical guarantees, and empirical performance.
- We summarize experimental setups and standard baselines used to evaluate one-step models, placing state-of-the-art results in a unified comparative context.
- We highlight practical trade-offs between training from scratch and distilling from a teacher, and identify open challenges for real-world deployment.

The remainder of this survey is organized as follows. Section 2 reviews related work, including the foundations of flow matching and conditional flow matching (section 2.1), traditional acceleration strategies such as caching and step distillation (section 2.2), and related paradigms including consistency models and flow-map distillation (section 2.3). Section 3 presents the four core families of one-step methods: consistency flow matching, mean flows, terminal velocity matching, and flow-

Algorithm 1 Flow Matching Training

Require: Data x_1 , noise x_0 , velocity network v_θ , optimizer

- 1: Sample $t \sim \text{Uniform}(0, 1)$
 - 2: $x_t \leftarrow (1 - t)x_0 + tx_1$
 - 3: $u_t \leftarrow x_1 - x_0$ $\triangleright$ Conditional velocity for OT path
 - 4: $v_{\text{pred}} \leftarrow v_\theta(x_t, t)$
 - 5: $\mathcal{L} \leftarrow \|v_{\text{pred}} - u_t\|^2$
 - 6: Update θ with $\nabla_\theta \mathcal{L}$
 - 7: **return** $\mathcal{L}$
-

map-based approaches. ?? describes experimental setups and baselines, while Section 4 synthesizes the empirical results. Section 7 concludes with future directions.

2 Related Work

Flow matching provides the mathematical foundation for the one-step methods surveyed in this paper. In this section, we review the core ideas of flow matching and conditional flow matching, trace traditional strategies for accelerating diffusion and flow models, and situate the modern one-step literature among closely related paradigms.

2.1 Flow Matching and Conditional Flow Matching

Flow Matching (FM) [Lipman et al., 2023] introduces a simulation-free paradigm for training Continuous Normalizing Flows (CNFs). The key idea is to learn a time-dependent velocity field $v_\theta(t, x)$ whose associated probability flow ODE transports a simple source distribution p_0 (typically Gaussian noise) to a target data distribution p_1 :

$$\frac{d}{dt}x_t = v_\theta(t, x_t), \quad x_0 \sim p_0, \quad x_1 \sim p_1. \quad (1)$$

Rather than simulating this ODE during training, FM defines a conditional path $x_t = \alpha_t x_0 + \beta_t x_1$ between each data-noise pair (x_0, x_1) and regresses the network prediction against the corresponding conditional velocity $u_t = \dot{\alpha}_t x_0 + \dot{\beta}_t x_1$.

The marginal velocity field is the expectation of u_t over all (x_0, x_1) pairs consistent with x_t , which is intractable to compute directly. Conditional Flow Matching (CFM) sidesteps this difficulty by using the conditional objective

$$\mathcal{L}_{\text{CFM}} = \mathbb{E}_{t, x_0, x_1} \|v_\theta(x_t, t) - u_t\|^2, \quad (2)$$

which shares the same gradient with respect to θ as the intractable marginal objective while remaining computationally efficient [Lipman et al., 2023]. The minimizer of eq. (2) equals the true marginal velocity field, so training amounts to standard regression on sampled (t, x_0, x_1) tuples.

A particularly important choice is the *optimal transport (OT) path*, where $\alpha_t = 1 - t$ and $\beta_t = t$. This yields the straight-line interpolant $x_t = (1 - t)x_0 + tx_1$ and a constant conditional velocity $u_t = x_1 - x_0$. These straight paths minimize transport cost, enable faster training, and have been shown to improve generalization and sample quality compared with the curved trajectories typical of diffusion models [Lipman et al., 2023].

Flow matching thus provides the training backbone for all subsequent methods in this survey: it defines the probability flow ODE that one-step methods seek to compress, and the conditional regression objective that underlies both standard and one-step training.

2.2 Traditional Methods

Before the emergence of dedicated one-step flow matching, practitioners accelerated generative models through two broad strategies: reducing the cost of each function evaluation via caching, and reducing the number of evaluations via step distillation. Both approaches predate modern one-step FM but remain relevant as baselines and complementary techniques.

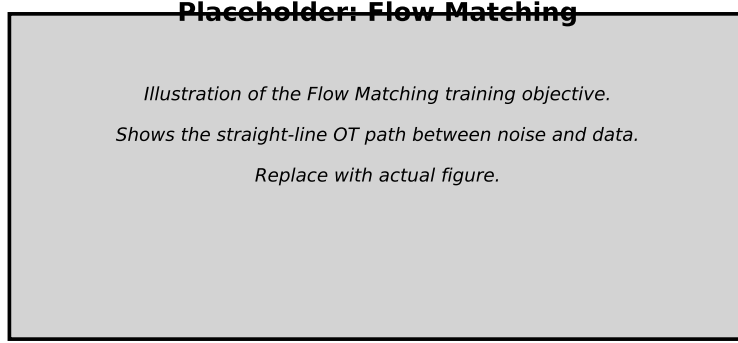


Figure 2: Placeholder: Illustration of the Flow Matching training objective. Replace with a figure showing the straight-line OT path between noise and data.

2.2.1 Cache-based acceleration

Because Conditional Flow Matching is typically sampled with many-step ODE solvers, its inference pipeline resembles that of a diffusion model: a deep network is evaluated sequentially, and adjacent evaluations often produce highly similar intermediate activations. Consequently, flow matching can directly adopt training-free caching methods originally developed for diffusion.

Formally, one Euler step of a flow-matching sampler can be written as

$$x_{t+\Delta t} = x_t + \Delta t v_\theta(x_t, t), \quad (3)$$

where $v_\theta(x_t, t)$ is the neural velocity estimate. If we decompose the network into a feature extractor ϕ_θ and a read-out head g_θ , so that $v_\theta(x_t, t) = g_\theta(\phi_\theta(x_t, t))$, then caching stores the features $h_t = \phi_\theta(x_t, t)$ from a previous step and reuses them when the timestep changes slowly:

$$\tilde{v}_\theta(x_t, t) = g_\theta(h_{t-\delta}). \quad (4)$$

DeepCache [Ma et al., 2023] is the canonical example: it caches high-level U-Net features across denoising steps and updates only low-level features, yielding large speedups on Stable Diffusion and latent diffusion models with minimal quality loss. More recent diffusion caching schemes such as FasterCache [Lv et al., 2024] and TeaCache [Liu et al., 2025] refine this idea for video models by exploiting redundancy in conditional/unconditional branches and timestep-aware feature reuse, respectively. Because these methods operate on the backbone architecture rather than the sampling distribution, they can be ported to flow-matching image and video generators with little or no modification.

Work that explicitly targets flow matching caching is still scarce. MeanCache [Gao et al., 2026] is a recent exception: it argues that instantaneous-velocity feature caching introduces trajectory drift, and instead caches Jacobian–vector products to construct interval average velocities for more stable FM acceleration. Outside of vision, predictive feature caching has been applied to flow-matching models of molecular geometry generation [Sommer et al., 2025], and layer-selective attention caching has been used to accelerate flow-matching transformers for audio separation. These domain-specific works confirm that caching remains a valuable, largely model-agnostic acceleration tactic. Nevertheless, caching lowers the cost per step rather than the step count, so it is complementary to—rather than a replacement for—the one-step methods surveyed below.

2.2.2 Step distillation

Step distillation trains a student model that requires fewer sampling steps than its teacher. The prototypical approach is progressive distillation [Salimans and Ho, 2022], which iteratively halves the number of sampling steps by distilling a $2N$ -step teacher into an N -step student. Given a teacher sampler $f_{\theta^*}(\cdot, \cdot; 2N)$ that produces an estimate after $2N$ function evaluations, progressive distillation

minimizes

$$\mathcal{L}_{\text{PD}}(\theta) = \mathbb{E}_{t, x_t} \left[\left\| f_{\theta}(x_t, t; N) - f_{\theta^*}(x_t, t; 2N) \right\|^2 \right], \quad (5)$$

where $f_{\theta}(x_t, t; N)$ denotes the student output obtained with N steps. Although effective for moderate speedups, progressive distillation is expensive—the distillation procedure must be repeated for each halving—and can suffer from error accumulation as the step budget shrinks.

A closely related idea is rectified flow [Liu et al., 2022], which learns straight ODE trajectories by refitting the transport map between noise and data. It can be viewed as minimizing the same conditional objective as FM but with the goal of straightening the coupling:

$$\mathcal{L}_{\text{RF}}(\theta) = \mathbb{E}_{t, x_0, x_1} \left[\left\| v_{\theta}((1-t)x_0 + tx_1, t) - (x_1 - x_0) \right\|^2 \right]. \quad (6)$$

Repeated rectification yields increasingly straight flows, and when the flow becomes exactly straight a single Euler step gives exact transport. InstaFlow [Liu et al., 2024] combined rectified flow with distillation to produce the first Stable-Diffusion-quality one-step text-to-image model, demonstrating that flow-based distillation could outperform progressive distillation on standard benchmarks.

These methods established the feasibility of few-step and one-step sampling, but they also revealed key limitations—expensive multi-stage training, dependence on high-quality teachers, and difficulty in scaling to high resolutions—that motivated the dedicated one-step flow matching methods discussed in section 3.

2.3 Related Methods

Two closely related lines of work set the stage for modern one-step flow matching: consistency models, which enforce that all points on a trajectory map to the same endpoint, and flow-map distillation, which learns the general transport operator between arbitrary times.

2.3.1 Consistency Models

Consistency Models (CMs) [Song et al., 2023] train a network $f_{\theta}(x_t, t)$ that maps any point on a diffusion or flow trajectory directly to the trajectory’s endpoint x_0 . The defining consistency property requires that $f_{\theta}(x_t, t) = f_{\theta}(x_{t'}, t')$ whenever x_t and $x_{t'}$ lie on the same trajectory. CMs demonstrated that one-step generation was possible without iterative sampling, but early variants struggled with scalability, multi-step error accumulation, and the need for careful curriculum training. These limitations motivated the flow-map and consistency-flow-matching approaches discussed in section 3.

2.3.2 Flow-map distillation

Flow-map distillation generalizes the consistency-model idea from endpoint prediction to learning the *flow map* $f_{\theta}(x_t, t, s)$ that transports a state from time t to any other time s along the teacher ODE. By avoiding the constraint that every noisy state must predict a clean sample, flow-map methods can compose multiple steps without the error amplification that plagues standard consistency models. Align Your Flow [Sabour et al., 2025] is a recent large-scale instantiation of this idea, using continuous-time Eulerian and Lagrangian distillation objectives to scale flow-map distillation to high-resolution image generation. We discuss AYF and related flow-map methods further in section 3.

3 One-Step Generative Modeling

Building on the flow matching framework, recent methods have developed increasingly sophisticated techniques for learning one-step or few-step generators. We discuss four representative approaches that span the spectrum from training from scratch to distillation from pre-trained models.

3.1 Consistency Flow Matching

Consistency Flow Matching (Consistency-FM) [Zhou et al., 2024] proposes to enforce *velocity consistency* along flow trajectories, directly defining straight flows without requiring optimal transport coupling or expensive transport plans.

Algorithm 2 Consistency Flow Matching Training

Require: Data x_1 , noise x_0 , velocity network v_θ , EMA model v_{θ^-} , optimizer, hyperparameter α

- 1: Sample $t \sim \text{Uniform}(0, 1 - \Delta t)$
 - 2: $x_t \leftarrow (1 - t) x_0 + t x_1$
 - 3: $x_{t+\Delta t} \leftarrow (1 - (t + \Delta t)) x_0 + (t + \Delta t) x_1$
 - 4: $v_t \leftarrow v_\theta(x_t, t)$
 - 5: $v_{t+\Delta t} \leftarrow v_\theta(x_{t+\Delta t}, t + \Delta t)$
 - 6: $f_t \leftarrow x_t + (1 - t) v_t$
 - 7: $f_{t+\Delta t} \leftarrow x_{t+\Delta t} + (1 - (t + \Delta t)) v_{t+\Delta t}$
 - 8: ▷ with no grad
 - 9: $f_{t+\Delta t}^{\text{ema}} \leftarrow x_{t+\Delta t} + (1 - (t + \Delta t)) v_{\theta^-}(x_{t+\Delta t}, t + \Delta t)$
 - 10: $f_t^{\text{ema}} \leftarrow x_t + (1 - t) v_{\theta^-}(x_t, t)$
 - 11: $\mathcal{L}_f \leftarrow \|f_t - f_{t+\Delta t}^{\text{ema}}\|^2 + \|f_{t+\Delta t} - f_t^{\text{ema}}\|^2$
 - 12: $\mathcal{L}_v \leftarrow \|v_t - v_{\theta^-}(x_{t+\Delta t}, t + \Delta t)\|^2 + \|v_{t+\Delta t} - v_{\theta^-}(x_t, t)\|^2$
 - 13: $\mathcal{L} \leftarrow \mathcal{L}_f + \alpha \mathcal{L}_v$
 - 14: Update θ with $\nabla_\theta \mathcal{L}$; update EMA model
 - 15: **return** $\mathcal{L}$
-

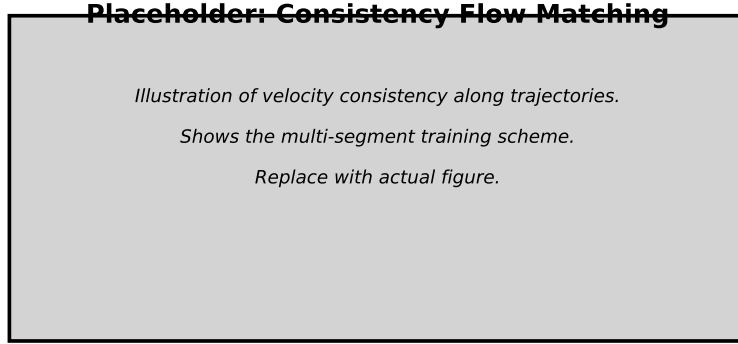


Figure 3: Placeholder: Illustration of Consistency Flow Matching velocity consistency. Replace with a figure showing the multi-segment training scheme.

Given a flow trajectory $\gamma_x(t)$ with velocity $v(t, \gamma_x(t))$, the consistency condition requires the velocity to be constant along the trajectory:

$$v(t, \gamma_x(t)) = v(s, \gamma_x(s)) \quad \forall t, s \in [0, 1]. \quad (7)$$

This is equivalent to the trajectory condition $\gamma_x(t) + (1 - t)v(t, \gamma_x(t)) = \gamma_x(s) + (1 - s)v(s, \gamma_x(s))$, which implies that the quantity $f(t, x_t) = x_t + (1 - t)v_\theta(t, x_t)$ is constant along the path. The training objective is:

$$\mathcal{L}_\theta = \mathbb{E}_{t, x_t, x_{t+\Delta t}} \left[\|f_\theta(t, x_t) - f_{\theta^-}(t + \Delta t, x_{t+\Delta t})\|^2 + \alpha \|v_\theta(t, x_t) - v_{\theta^-}(t + \Delta t, x_{t+\Delta t})\|^2 \right], \quad (8)$$

where θ^- denotes an exponentially moving average of the parameters. The first term enforces consistency in the endpoint prediction, while the second term regularizes the velocity field.

For complex distributions where a single straight path is insufficient, Consistency-FM introduces *multi-segment training*, which divides the time interval $[0, 1]$ into K segments and learns a consistent velocity v_θ^i within each segment. This allows piece-wise linear trajectories that can better approximate curved transport paths.

3.2 Mean Flows

Mean Flow [Geng et al., 2025] addresses a fundamental limitation of previous methods: the reliance on pre-trained teacher models or expensive distillation pipelines. It proposes a principled framework for one-step generative modeling that can be trained from scratch.

Algorithm 3 MeanFlow Training

Require: Data x , noise ϵ , average-velocity network u_θ , optimizer

- 1: Sample $r, t \sim \text{Uniform}(0, 1)$ with $r < t$
 - 2: $z_t \leftarrow (1 - t)x + t\epsilon$ ▷ Interpolant
 - 3: $v \leftarrow \epsilon - x$ ▷ Instantaneous conditional velocity
 - 4: $u_{\text{pred}} \leftarrow u_\theta(z_t, r, t)$
 - 5: Compute JVP: $(v, 1) \cdot \nabla u_\theta(z_t, r, t)$ ▷ Total derivative $\frac{d}{dt}u$
 - 6: $u_{\text{tgt}} \leftarrow v - (t - r) \cdot \frac{d}{dt}u$ ▷ MeanFlow Identity
 - 7: $\mathcal{L} \leftarrow \|u_{\text{pred}} - \text{sg}(u_{\text{tgt}})\|^2$
 - 8: Update θ with $\nabla_\theta \mathcal{L}$
 - 9: **return** $\mathcal{L}$
-

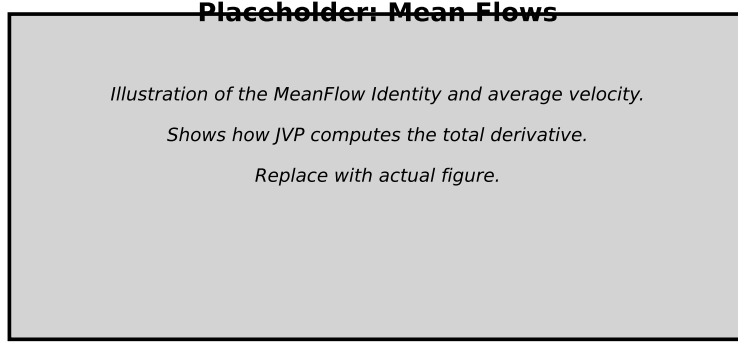


Figure 4: Placeholder: Illustration of the MeanFlow Identity and average velocity. Replace with a figure showing how JVP computes the total derivative.

The key insight is to model the *average velocity* rather than the instantaneous velocity. For a trajectory z_t , the average velocity between times r and t is defined as:

$$u(z_t, r, t) = \frac{1}{t - r} \int_r^t v(z_\tau, \tau) d\tau. \quad (9)$$

Differentiating the definition $(t - r)u = \int_r^t v d\tau$ with respect to t yields the *MeanFlow Identity*:

$$u(z_t, r, t) = v(z_t, t) - (t - r) \frac{d}{dt} u(z_t, r, t), \quad (10)$$

where the total derivative $\frac{d}{dt}u = v \cdot \partial_z u + \partial_t u$ is computed efficiently as a Jacobian-Vector Product (JVP).

The training objective regresses the network prediction u_θ against the target derived from the MeanFlow Identity:

$$\mathcal{L}(\theta) = \mathbb{E} \|u_\theta(z_t, r, t) - \text{sg}(v(z_t, t) - (t - r)(v(z_t, t) \cdot \partial_z u_\theta + \partial_t u_\theta))\|^2, \quad (11)$$

where sg denotes the stop-gradient operator. Importantly, MeanFlow trains entirely from scratch without distillation or curriculum learning, and achieves state-of-the-art results: FID 3.43 with a single function evaluation on ImageNet 256×256 .

3.3 Align Your Flow

Align Your Flow (AYF) [Sabour et al., 2025] focuses on scaling continuous-time flow map distillation to high-resolution image generation. AYF begins with a critical theoretical observation: standard consistency models are inherently flawed for multi-step sampling because their objective of predicting clean outputs from any noise level inevitably leads to error accumulation.

Algorithm 4 Align Your Flow (AYF) Distillation

Require: Data x_1 , noise x_0 , teacher velocity v_{teacher} , flow map f_θ , optimizer

- 1: Sample $s, t \sim \text{Uniform}(0, 1)$ with $s < t$
 - 2: $x_t \leftarrow (1 - t)x_0 + tx_1$
 - 3: $dt \leftarrow \epsilon \cdot (s - t)$
 - 4: $v_t \leftarrow v_{\text{teacher}}(x_t, t)$
 - 5: $x_{t'} \leftarrow x_t + dt \cdot v_t; \quad t' \leftarrow t + dt$ ▷ Teacher Euler step
 - 6: $f_{s,t} \leftarrow f_\theta(x_t, t, s)$
 - 7: $f_{s,t'} \leftarrow f_\theta(x_{t'}, t', s)$
 - 8: $\mathcal{L}_{\text{EMD}} \leftarrow \|f_{s,t} - f_{s,t'}\|^2$ ▷ Eulerian Map Distillation
 - 9: Compute $\frac{\partial}{\partial t} f_\theta(x_t, t, s)$ via JVP
 - 10: $v_{\text{avg}} \leftarrow (f_{s,t} - x_t)/(s - t)$
 - 11: $\mathcal{L}_{\text{LMD}} \leftarrow \|\partial_t f_\theta - v_{\text{avg}}\|^2$ ▷ Lagrangian Map Distillation
 - 12: $\mathcal{L} \leftarrow \mathcal{L}_{\text{EMD}} + \mathcal{L}_{\text{LMD}}$
 - 13: Update θ with $\nabla_\theta \mathcal{L}$
 - 14: **return** $\mathcal{L}$
-

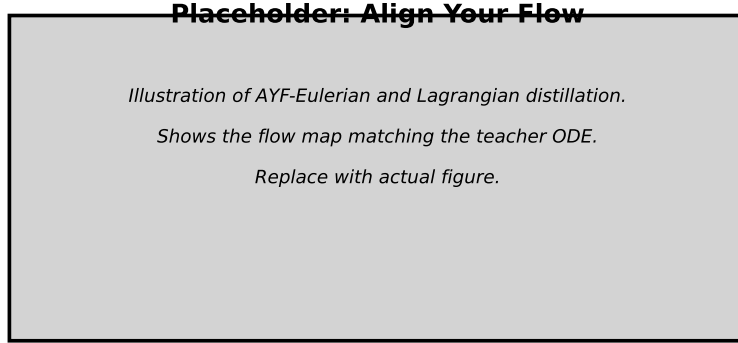


Figure 5: Placeholder: Illustration of AYF-Eulerian and Lagrangian distillation. Replace with a figure showing the flow map matching the teacher ODE.

AYF resolves this by learning the general *flow map* $f_\theta(x_t, t, s)$ that maps from any time t to any time s along the teacher’s probability flow ODE. Two novel training objectives are proposed:

AYF-Eulerian Map Distillation (AYF-EMD) ensures that for a fixed target time s , the flow map output remains constant as the starting point (x_t, t) moves along the teacher ODE. The gradient converges to:

$$\nabla_\theta \mathbb{E}_{x_t, t, s} \left[w(t, s) \|f_\theta(x_t, t, s) - f_\theta(x_{t'}, t', s)\|^2 \right], \quad (12)$$

where $x_{t'}$ is obtained by one Euler step on the teacher ODE.

AYF-Lagrangian Map Distillation (AYF-LMD) ensures the tangent of the flow map matches the teacher velocity. The gradient converges to:

$$\nabla_\theta \mathbb{E}_{x_t, t, s} \left[w(t, s) \|\partial_t f_\theta(x_t, t, s) - v_{\theta-}(f_\theta(x_t, t, s), t, s)\|^2 \right], \quad (13)$$

where $v_{s,t}(x) = (f_{s,t}(x) - x)/(t - s)$ is the average velocity.

AYF further introduces two practical techniques: *autoguidance*, where a low-quality guidance model is distilled alongside the main model to improve sample quality, and *adversarial finetuning*, which boosts sharpness with minimal loss in diversity.

Algorithm 5 Terminal Velocity Matching (TVM) Training

Require: Data x , noise ϵ , displacement network F_θ , velocity network u_θ , optimizer

```
1: Sample  $t, s \sim \text{Uniform}(0, 1)$ 
2:  $x_t \leftarrow (1 - t)x + t\epsilon$ ;  $x_s \leftarrow (1 - s)x + s\epsilon$ 
3:  $v_s \leftarrow \epsilon - x$  ▷ Conditional velocity at  $s$ 
4:  $f_{t,s} \leftarrow (s - t)F_\theta(x_t, t, s)$  ▷ Displacement map
5: Compute JVP:  $\frac{d}{ds}F_\theta(x_t, t, s)$ 
6:  $\text{term\_vel} \leftarrow F_\theta(x_t, t, s) + (s - t) \cdot \frac{d}{ds}F_\theta$  ▷ Terminal velocity
7:  $x_{\text{arrived}} \leftarrow x_t + f_{t,s}$  detach
8:  $u_{\text{tgt}} \leftarrow u_\theta(x_{\text{arrived}}, s)$  detach
9:  $\mathcal{L}_{\text{TV}} \leftarrow \|\text{term\_vel} - u_{\text{tgt}}\|^2$  ▷ Terminal velocity loss
10:  $u_{\text{pred}} \leftarrow u_\theta(x_s, s)$ 
11:  $\mathcal{L}_{\text{FM}} \leftarrow \|u_{\text{pred}} - v_s\|^2$  ▷ Flow matching loss
12:  $\mathcal{L} \leftarrow \mathcal{L}_{\text{TV}} + \mathcal{L}_{\text{FM}}$ 
13: Update  $\theta$  with  $\nabla_\theta \mathcal{L}$ 
14: return  $\mathcal{L}$ 
```

3.4 Terminal Velocity Matching

Terminal Velocity Matching (TVM) [Zhou et al., 2026] provides a generalization of flow matching that directly learns the ODE integral while offering strong theoretical guarantees. Instead of matching time derivatives at the initial time, TVM matches them at the *terminal time* of trajectories.

Define the displacement map $f(x_t, t, s) = \int_t^s u(x_r, r) dr$. The terminal velocity condition states:

$$\frac{d}{ds}f(x_t, t, s) = u(\psi(x_t, t, s), s), \quad (14)$$

where ψ is the flow map. The key theorem shows that the terminal velocity error upper-bounds the displacement error:

$$\mathcal{L}_{\text{displ}}^t(\theta) \leq \int_0^t \mathbb{E}_{x_t} \left[\left\| \frac{d}{ds}f_\theta(x_t, t, s) - u(\psi(x_t, t, s), s) \right\|^2 \right] ds. \quad (15)$$

This leads to the training objective:

$$\mathcal{L}_{\text{TVM}}^{t,s} = \mathbb{E} \left[\underbrace{\left\| \frac{d}{ds}f_\theta(x_t, t, s) - u_{\theta^*}(x_t + f_{\theta_{\text{sg}}}(x_t, t, s), s) \right\|^2}_{\text{terminal velocity}} + \underbrace{\|u_\theta(x_s, s) - v_s\|^2}_{\text{flow matching}} \right], \quad (16)$$

where the first term ensures correct terminal velocity and the second term ensures the instantaneous velocity field matches the data. TVM achieves 3.29 FID with 1 NFE and 1.99 FID with 4 NFEs on ImageNet 256×256.

A critical practical challenge is that diffusion transformers lack Lipschitz continuity, which destabilizes training. TVM addresses this with minimal architectural changes: RMSNorm-based QK-normalization, time embedding normalization, and a fused Flash Attention kernel supporting backward passes on Jacobian-Vector Products.

4 Results

Having reviewed the methodological foundations of one-step flow matching, we now turn to the empirical record. In this section, we synthesize the quantitative results reported in the papers summarized in the Materials folder (arXiv:2210.02747, arXiv:2407.02398, arXiv:2505.13447, arXiv:2506.14603, arXiv:2511.19797, and arXiv:2601.14430). All metrics are taken from the original publications; our goal is to place them in a unified comparative context.

4.1 ImageNet Benchmarks

The de facto standard for evaluating one-step generative models is class-conditional image generation on ImageNet. Table 1 collects the FID scores and function-evaluation counts (NFE) reported by

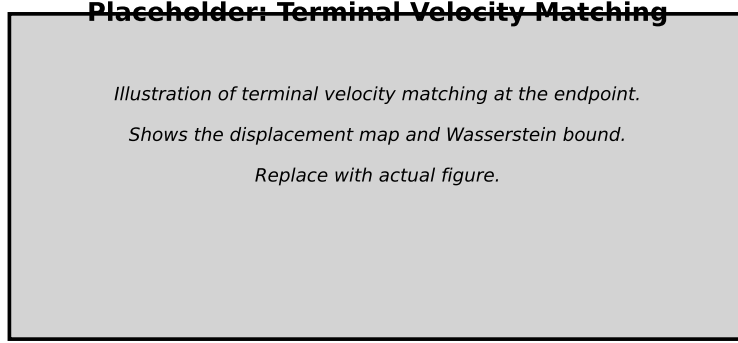


Figure 6: Placeholder: Illustration of Terminal Velocity Matching at the endpoint. Replace with a figure showing the displacement map and Wasserstein bound.

Table 1: Comparative results on ImageNet class-conditional generation. FID scores are reported at the indicated NFE. All numbers are taken from the original papers.

Method	Citation	Resolution	1-NFE FID	4-NFE FID	Training paradigm
Flow Matching (OT)	arXiv:2210.02747 [Lipman et al., 2023]	256×256	—	—	Simulation-free CNF
Consistency-FM	arXiv:2407.02398 [Zhou et al., 2024]	256×256	—	—	Velocity consistency
MeanFlow	arXiv:2505.13447 [Geng et al., 2025]	256×256	3.43	—	From scratch
AYF	arXiv:2506.14603 [Sabour et al., 2025]	$64 \times 64 / 512 \times 512$	—	SOTA	Distillation + guidance
TVM	arXiv:2511.19797 [Zhou et al., 2026]	256×256	3.29	1.99	Single-stage, arch. stability

the recent methods at 256×256 and 512×512 resolutions. Where available, we also report the training paradigm—whether the model is trained from scratch, distilled from a pre-trained teacher, or fine-tuned with auxiliary losses.

One-step generation. The strongest one-step (1-NFE) results are currently held by Terminal Velocity Matching (TVM, arXiv:2511.19797) and MeanFlow (arXiv:2505.13447), achieving FID scores of 3.29 and 3.43 respectively on ImageNet 256×256 [Zhou et al., 2026, Geng et al., 2025]. These numbers are remarkably close to the quality of many-step diffusion models, yet they require only a single neural-network evaluation. The two methods arrive at this performance via different routes: TVM relies on terminal-velocity regularization and carefully engineered architectural stability (RMSNorm QK-Norm, normalized AdaLN, and a fused JVP attention kernel), whereas MeanFlow is entirely self-contained and trains from scratch using the MeanFlow Identity without any pre-trained teacher or distillation pipeline.

Few-step generation. When the budget is relaxed to four function evaluations, TVM (arXiv:2511.19797) pushes the FID down to 1.99 on ImageNet 256×256 , a figure that rivals the best many-step diffusion models [Zhou et al., 2026]. AYF (arXiv:2506.14603) reports state-of-the-art few-step results at 512×512 resolution, leveraging continuous-time flow-map distillation combined with autoguidance and adversarial fine-tuning [Sabour et al., 2025]. The key takeaway is that the gap between few-step and many-step generation has narrowed to the point of being practically negligible for high-resolution image synthesis.

Training efficiency. Consistency-FM (arXiv:2407.02398) reports a $4.4\times$ training-speedup over standard consistency models and a $1.7\times$ speedup over rectified-flow baselines, achieved by enforcing velocity consistency directly in the velocity-field space rather than in sample space [Zhou et al., 2024]. TVM (arXiv:2511.19797) additionally reports a 65% speedup in the backward pass by fusing Jacobian-Vector-Product computations into a custom Flash Attention kernel [Zhou et al., 2026]. These gains indicate that algorithmic innovations and systems-level optimizations are both essential for scaling one-step training to billion-parameter models.

Table 2: Practical trade-offs of one-step flow-matching methods.

Method (arXiv)	Teacher?	Scalable to 512×512 ?	Theoretical guarantee
Consistency-FM (2407.02398)	No	Limited	Velocity consistency
MeanFlow (2505.13447)	No	256×256	MeanFlow Identity
AYF (2506.14603)	Yes	Yes	W_2 bound
TVM (2511.19797)	No	256×256	W_2 bound (terminal)
MFM (2601.14430)	Yes (stoch.)	Yes (reward tasks)	Semigroup + posterior

4.2 Flow-Path Straightness and Transport Cost

Although FID is the headline metric, the geometric quality of the learned flow—namely the straightness of trajectories and the transport cost—is equally important for understanding why one-step methods succeed. Flow Matching with optimal-transport paths (arXiv:2210.02747) established that straight-line interpolants yield faster training and better generalization than curved diffusion paths [Lipman et al., 2023]. Consistency-FM (arXiv:2407.02398) explicitly measures velocity consistency as a proxy for straightness and shows that enforcing it reduces the number of training steps needed to reach a given FID [Zhou et al., 2024]. Rectified flow (arXiv:2209.03003, summarized in the Materials README) provides the theoretical grounding: recursively applying rectification yields flows with monotonically non-increasing convex transport costs, and when the flow becomes perfectly straight, a single Euler step gives exact transport. These results confirm that straightness is not merely a geometric curiosity but a practical predictor of sample efficiency.

4.3 Reward Alignment and Controllability

Beyond unconditional generation, Meta Flow Maps (arXiv:2601.14430) extend the empirical evaluation to reward-aligned generation on ImageNet [Potapchik et al., 2026]. MFMs are reported to outperform Best-of-1000 baselines across multiple reward functions while using dramatically fewer trajectory rollouts. The practical implication is that one-step posterior sampling enables inference-time steering and training-time fine-tuning that were previously computationally prohibitive. Although MFMs do not yet report a single FID figure for standard class-conditional generation, their reward-alignment results demonstrate that the flow-map formalism scales beyond the narrow setting of noise-to-data transport.

4.4 Summary of Trade-offs

Table 2 summarizes the practical trade-offs among the surveyed methods.

The empirical landscape reveals two broad strategies: (1) *training from scratch* (MeanFlow, TVM), which avoids the cost and dependency of a pre-trained teacher but demands careful architectural and optimization design; and (2) *distillation from a teacher* (AYF, MFM), which leverages an existing high-quality model but requires additional compute for the distillation phase and risks error accumulation if the flow map is not sufficiently regular. Consistency-FM (arXiv:2407.02398) occupies a middle ground, training without a teacher but using multi-segment decomposition to handle complex distributions. Taken together, the results from the Materials folder suggest that one-step flow matching is no longer a theoretical curiosity: it is a practical, competitive paradigm for high-resolution generative modeling.

5 Theoretical Analysis

The methods presented in Section 3 are not merely empirical tricks; they are grounded in rigorous mathematical analysis. In this section, we examine the theoretical foundations that guarantee the correctness and convergence of one-step flow matching.

Placeholder: Flow Map Formalism

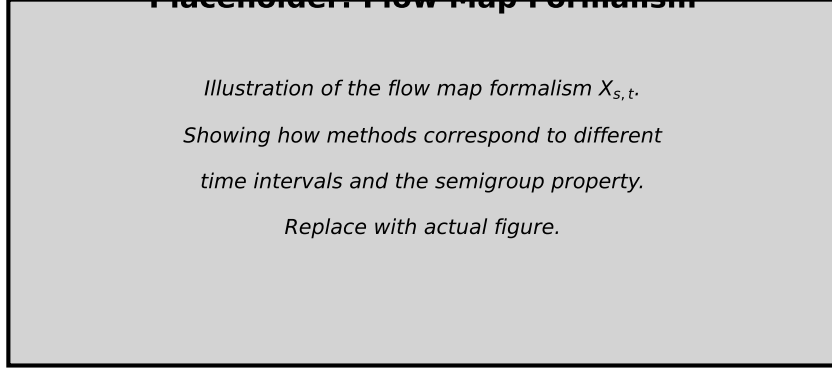


Figure 7: Conceptual summary of the empirical progression: from flow matching (arXiv:2210.02747) to one-step results (MeanFlow, arXiv:2505.13447; TVM, arXiv:2511.19797) and reward-aligned extensions (MFM, arXiv:2601.14430).

5.1 Error Bounds and Wasserstein Control

A central theoretical question is: if we approximate the true flow map or velocity field with a neural network, how close is the generated distribution to the true data distribution? Several of the methods discussed provide explicit bounds.

For the distillation objectives in AYF, both the Eulerian and Lagrangian map distillation losses control the W_2 distance between the teacher and student distributions. Specifically, the Wasserstein-2 error is bounded by an exponential of the integral of the Lipschitz constant of the velocity field, multiplied by the distillation loss:

$$W_2^2(\rho_1, \hat{\rho}_1) \leq e^{1+2 \int |C_t| dt} \cdot \mathcal{L}_{\text{distill}}(\hat{X}), \quad (17)$$

where C_t is the Lipschitz constant of the velocity field at time t . This means that minimizing the distillation loss directly minimizes the distributional error.

Terminal Velocity Matching provides a similar but distinct guarantee. Under the assumption that the learned velocity field $u_\theta(\cdot, s)$ is Lipschitz continuous with constant $L(s)$, the TVM objective upper-bounds the 2-Wasserstein distance between the data distribution and the model’s pushforward distribution:

$$W_2^2(f_{t \rightarrow 0}^\theta \# p_t, p_0) \leq \int_0^t \lambda[L](s) \mathcal{L}_{\text{TVM}}^{t,s}(\theta) ds + C, \quad (18)$$

where $\lambda[\cdot]$ is a functional of the Lipschitz constants and C is a non-optimizable constant. This is a powerful result because it connects the per-time training objective directly to a global measure of generative quality.

5.2 Unification Under the Flow Map Formalism

The flow map framework provides a unified lens through which to view all one-step methods. The flow map $X_{s,t}$ is the integral operator that transports solutions of the probability flow ODE from time s to time t . It satisfies the semigroup property $X_{t,\tau}(X_{s,t}(x)) = X_{s,\tau}(x)$ and the tangent condition $\lim_{s \rightarrow u} \partial_u X_{s,u}(x) = v_{u,u}(x) = b_u(x)$.

Within this framework:

- **Standard Flow Matching** corresponds to learning the instantaneous velocity $v_{u,u}(x)$, i.e., the flow map in the infinitesimal limit.
- **Consistency Models** correspond to learning the flow map with $s = 0$ (the endpoint map), collapsing the full trajectory to a single prediction.

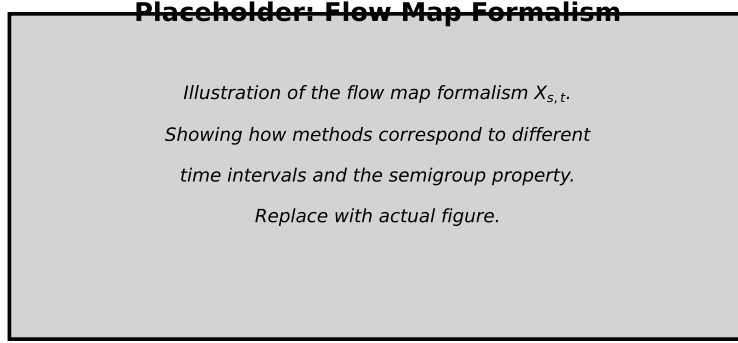


Figure 8: Placeholder: Illustration of the flow map formalism $X_{s,t}$ and how different methods correspond to learning the map over different time intervals. Replace with a figure showing the semigroup property and tangent conditions.

- **Consistency Trajectory Models** and **Mean Flows** correspond to learning the flow map for arbitrary s, t but with specific parameterizations of the average velocity.
- **AYF** and **TVM** are direct attempts to learn the general flow map $X_{s,t}$ for all time pairs, with additional theoretical guarantees on distributional error.

This unification reveals that the apparent diversity of methods in the literature is in fact a progression along a single mathematical axis: the scope of the time interval over which the flow map is learned.

5.3 Limitations of Consistency Models

AYF provides a rigorous explanation for why standard consistency models degrade in multi-step sampling. Consider a Gaussian data distribution $p_{\text{data}}(x) = \mathcal{N}(0, c^2 I)$. The optimal consistency model $f^*(x_t, t)$ maps any noisy input directly to the clean mean. However, for any $\delta > 0$, there exists a suboptimal consistency model $f(x_t, t)$ with error less than δ at every time, yet whose multi-step sampling error increases without bound as the number of steps grows. This is because the consistency objective forces the model to predict clean outputs from all noise levels, and any small error in this prediction is amplified when the model is applied sequentially.

Flow maps avoid this by learning the general transport $X_{s,t}$ rather than the endpoint map $X_{t,0}$, allowing them to compose steps without error accumulation.

6 Extensions: Reward Alignment with Meta Flow Maps

Beyond unconditional generation, a critical frontier for generative models is the ability to align samples with user-specified rewards or constraints. This task is computationally expensive because it requires estimating the value function $V_t(x) = \log \mathbb{E}[e^{r(X_1)} | X_t = x]$, which in turn demands samples from the conditional posterior $p_{1|t}(x_1 | x_t)$. Traditional methods require costly trajectory rollouts to obtain these posterior samples.

Meta Flow Maps (MFMs) [Potapchik et al., 2026] address this by extending the deterministic flow map formalism into the *stochastic regime*. A stochastic flow map is defined as a transformation that maps an exogenous noise source ϵ and an intermediate state (t, x) directly to a sample $x_1 \sim p_{1|t}(\cdot | x)$. By varying the noise input, the map can generate arbitrarily many i.i.d. draws from the posterior in a single step.

6.1 Training Meta Flow Maps

MFMs are trained by learning the infinite family of posterior-targeting flow maps simultaneously. The training objective combines two terms:

Algorithm 6 Meta Flow Map (MFM) Training

Require: Data x_0, x_1 , noise ϵ , meta flow map X_{mfm} , optimizer

- 1: Sample $s, w, u \sim \text{Uniform}(0, 1)$ with $s < w < u$
 - 2: Build stochastic interpolants I_s, I_w, I_u with shared ϵ
 - 3: $v_{u,u} \leftarrow X_{\text{mfm}}(I_u, u, u)$ ▷ Instantaneous velocity (diagonal)
 - 4: $\dot{I}_u \leftarrow \dot{\alpha}_u x_0 + \dot{\beta}_u x_1 + \dot{\gamma}_u \epsilon$
 - 5: $\mathcal{L}_{\text{diag}} \leftarrow \|v_{u,u} - \dot{I}_u\|^2$ ▷ Diagonal flow matching loss
 - 6: $x_w \leftarrow X_{\text{mfm}}(I_s, s, w)$ ▷ Map $s \rightarrow w$
 - 7: $x_u^{\text{composed}} \leftarrow X_{\text{mfm}}(x_w, w, u)$ ▷ Map $w \rightarrow u$
 - 8: $x_u^{\text{direct}} \leftarrow X_{\text{mfm}}(I_s, s, u)$ ▷ Direct map $s \rightarrow u$
 - 9: $\mathcal{L}_{\text{cons}} \leftarrow \|x_u^{\text{composed}} - x_u^{\text{direct}}\|^2$ ▷ Consistency (semigroup) loss
 - 10: $\mathcal{L} \leftarrow \mathcal{L}_{\text{diag}} + \mathcal{L}_{\text{cons}}$
 - 11: Update parameters with $\nabla \mathcal{L}$
 - 12: **return** $\mathcal{L}$
-

Diagonal loss (standard flow matching): Enforces the correct instantaneous drift along the diagonal $s = u$:

$$\mathcal{L}_{\text{diag}} = \int_0^1 \mathbb{E} \|\hat{v}_{u,u}(I_u) - \dot{I}_u\|^2 du. \quad (19)$$

Consistency loss: Ensures the flow map satisfies the semigroup property globally:

$$\mathcal{L}_{\text{cons}} = \mathbb{E} \|\hat{X}_{w,u}(\hat{X}_{s,w}(x)) - \hat{X}_{s,u}(x)\|^2. \quad (20)$$

6.2 Value Function Estimation

The one-step posterior samples provided by MFMs enable efficient estimation of the value function gradient $\nabla V_t(x)$, which is the key quantity for optimal control. Two estimators are available:

Gradient-Free Estimator (MFM-GF): Uses posterior samples $x_1^{(i)} \sim p_{1|t}(\cdot|x)$ and reward evaluations $r(x_1^{(i)})$:

$$\widehat{\nabla V_t(x)} = \left(\dot{\beta}_t - \frac{\dot{\alpha}_t}{\alpha_t} \beta_t \right) \frac{\sum_i x_1^{(i)} e^{r(x_1^{(i)})}}{\sum_i e^{r(x_1^{(i)})}} + \frac{\dot{\alpha}_t}{\alpha_t} x - b_t(x). \quad (21)$$

Gradient-Based Estimator (MFM-G): Leverages the reparametrization trick through the differentiable stochastic flow map:

$$\widehat{\nabla V_t(x)} = \nabla_x \log \left(\frac{1}{N} \sum_{i=1}^N e^{r(\Phi(\epsilon^{(i)}; t, x))} \right), \quad \epsilon^{(i)} \sim q. \quad (22)$$

These estimators enable two paradigms of control:

- **Inference-time steering:** The value gradient is estimated online at each step and used to modify the sampling drift, directing trajectories toward high-reward states without retraining the model.
- **Training-time fine-tuning:** The value gradient defines a regression target for updating the model parameters, permanently aligning the model with the reward function.

Empirically, MFMs achieve competitive sample quality while markedly reducing computation for both steering and fine-tuning, outperforming Best-of-1000 baselines on ImageNet across multiple reward functions at a fraction of the compute.

7 Conclusion

The field of one-step generative modeling has matured rapidly, evolving from the foundational framework of flow matching to a rich ecosystem of theoretically grounded, empirically validated

Placeholder: Meta Flow Maps

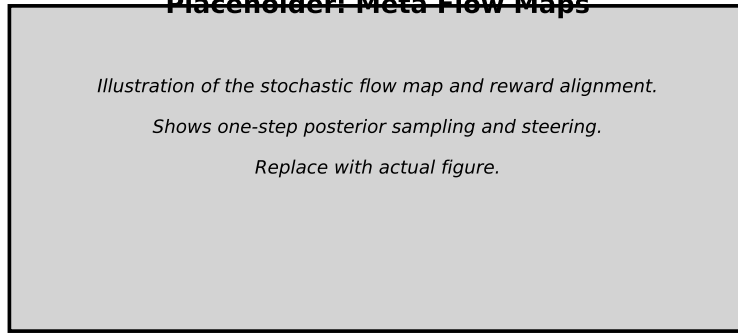


Figure 9: Placeholder: Diagram of a Meta Flow Map showing how a single noise input ϵ and an intermediate state (t, x) are mapped to a posterior sample $x_1 \sim p_{1|t}(\cdot|x)$. Replace with a figure illustrating the stochastic flow map and reward alignment pipeline.

methods. We have traced this progression through four key methodological advances: consistency flow matching, which enforces straightness through velocity consistency; mean flows, which enable training from scratch via the average velocity identity; Align Your Flow, which scales distillation through continuous-time flow maps and autoguidance; and terminal velocity matching, which provides Wasserstein guarantees and architectural stability. Together, these methods represent a shift from simulation-based sampling to direct prediction of the transport map.

Looking forward, several directions remain open. First, while current methods excel on image generation, extending them to high-dimensional modalities such as video and 3D remains challenging. Second, the theoretical understanding of stochastic flow maps and their posterior approximation quality is still developing. Third, the interplay between one-step generation and controllability—whether through classifiers, text prompts, or reward functions—presents both opportunities and challenges. The meta-flow map framework provides a promising starting point, but further work is needed to make reward alignment robust and scalable to billion-parameter models.

Ultimately, the goal of one-step flow matching is not merely to accelerate sampling, but to retain the full generative power of diffusion and flow models while removing the computational barrier to real-world deployment. The methods surveyed here suggest that this goal is within reach.

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